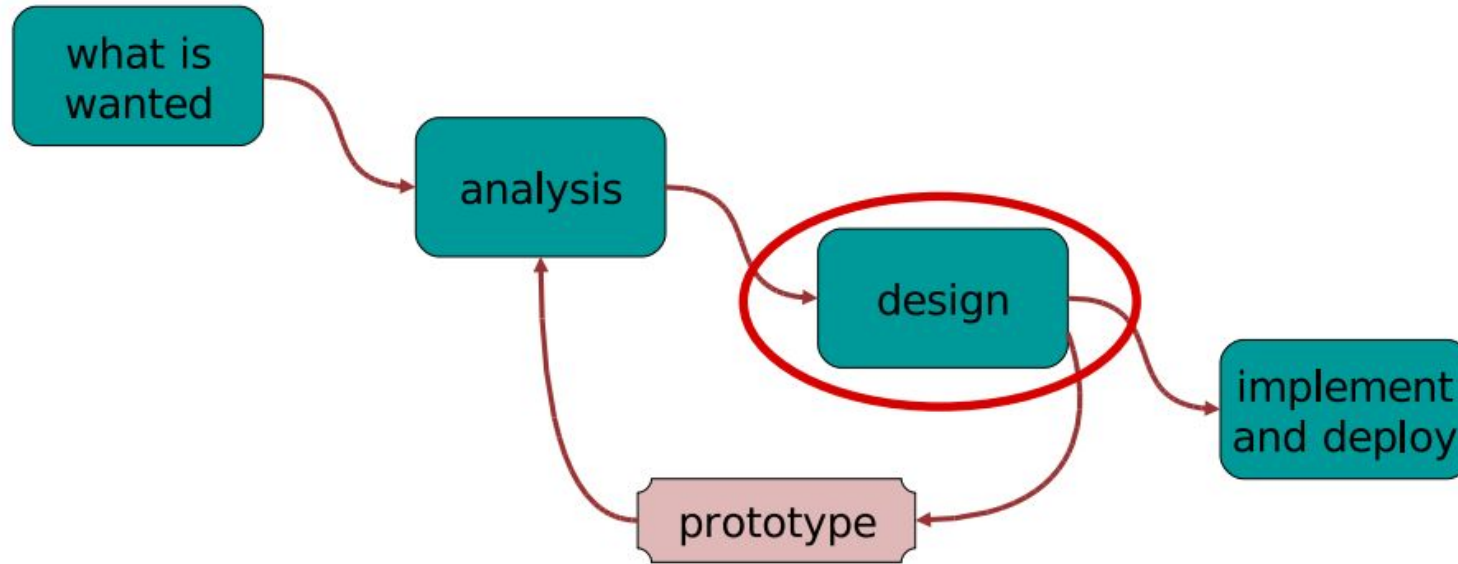


Conceptual design

Focus on Design

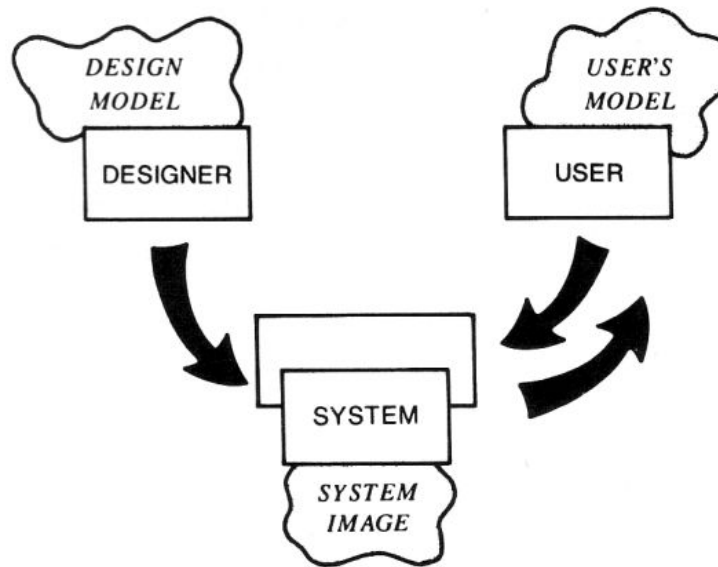


- How do we actually do the design?
- Temptation: start sketching windows, menus and buttons.
- But we can do better by starting from thinking about the user experience we want to provide.

Conceptual Models

A **conceptual model** is the designer's intended mental model for the user of the system: a set of ideas about how it is organized and operates.

Norman (1986) called this the *design model*:



Advantages of a Conceptual Model

- ▶ A conceptual model
 - ▶ is a starting point for interaction design
 - ▶ should help the user “figure it out”
- ▶ It helps design team:
 - ▶ Not to become narrowly focused early on
 - ▶ Ask questions about how the conceptual model will be understood by users
 - ▶ Establish a set of common terms they all understand and agree upon (a standard **lexicon** for the project)
 - ▶ Reduce the chance of misunderstandings and confusion arising later on

Example conceptual model (sketch)

Online library

metaphor information is organised as a physical card catalogue

concepts *item, book, periodical, issue, DVD, shelf-mark, user account, librarian, ...*

object relationships a book is a type of item;
periodicals contain issues

mappings *item* corresponds to a physical object;
shelf-mark to its physical location

functions *issue item, return item, search item*

function relationships issue before return for same
item; for different items, in parallel, ...

data new items added by typing data

Metaphors

- ▶ Interface metaphors combine familiar knowledge with new knowledge in a way that will help the user understand the product.
- ▶ Benefits:
 - ▶ make learning easier
 - ▶ enhances understanding of conceptual model
 - ▶ introduce innovation and widen accessibility
- ▶ Three steps to consider:
 1. understand functionality
 2. identify potential problem areas
 3. generate metaphors

Physical Design

- ▶ We may have lots or little choice:
 - ▶ a new special-purpose physical product, with our own choice of input/output features
 - ▶ new I/O mechanisms for existing device
 - ▶ new usage of existing mechanisms
 - ▶ standard device (e.g., PC) with standard mechanisms

Modes of interaction

Modes of interaction can be classified as:

- instructing** user tells system what to do, by typing commands, selecting menu options, pressing keys or buttons, speaking commands
- conversing** user has dialogue with system; typing questions and/or responses, or uses speech input/output
- manipulation** user interacts with physical or virtual objects, e.g., holding, moving, opening, closing; object is a *focus* of attention
- exploration** user moves through physical or virtual environment

Other possibilities and higher-level classifications exist, e.g., we may interact by **learning**, **problem solving**, **socializing**, **searching**, ...

Vending machines



Exercise: Interface for Robot Cleaner

Design an interface for controlling a robot vacuum cleaner.

1. Extend and deepen the task analysis for house cleaning given in the previous lecture, to consider:
 - ▶ individual tasks that performed by the robot
 - ▶ interactions necessary to control the robot
2. Propose a suitable conceptual model
3. Consider the physical design of the system
4. ... and interaction modes that would be appropriate for different tasks.
5. Justify your choices.

